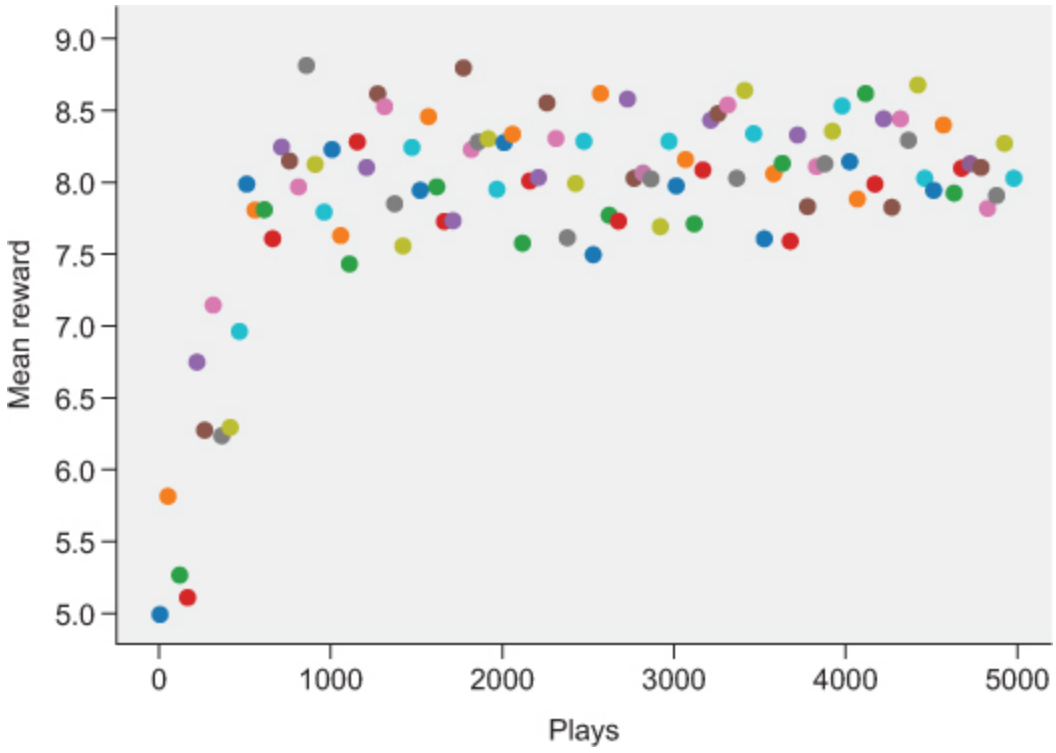


2.6. The Markov property

In our contextual bandit problem, our neural network led us to choose the best action given a state without reference to any other prior states. We just gave it the current state, and it produced the expected rewards for each possible action. This is an important property in reinforcement learning called the *Markov property*. A game (or any other control task) that exhibits the Markov property is said to be a *Markov decision process* (MDP). With an MDP, the current state alone contains enough information to choose optimal actions to maximize future rewards. Modeling a control task as an MDP is a key concept in reinforcement learning.

The MDP model simplifies an RL problem dramatically, as we do not need to take into account all previous states or actions—we don't need to have memory, we just need to analyze the present situation. Hence, we always attempt to model a problem as (at least approximately) a Markov decision processes. The card game Blackjack (also known as 21) is an MDP because we can play the game successfully just by knowing our current state (what cards we have, and the dealer's one face-up card).